Pyramakerz®

Unit 2

Lesson 2

Innovators in Action.

Less Effort, More Impact.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Calculate the amount of work and power involved in using a walker.
- Analyze how a design can reduce effort and improve efficiency.
- Compare different mobility aid designs and evaluate their strengths and weaknesses.
- Redesign their smart walker prototype to make it more ergonomic and accessible.
- Explain why inclusive design matters for people around the world.

STEAM SCOUTS

Challenge questions of the lesson

- What makes one walker easier to use than another?
- Can we design a mobility aid that uses less effort but helps more?
- How do power, efficiency, and comfort connect in real-life tools?
- Could someone in another country use the same design — why or why not?
- What if your walker design had to help someone on rough ground, or in a small apartment?

SEL Relation

- **Empathy & Perspective-Taking:** Students examine real-life struggles faced by people with mobility challenges. They imagine how those struggles vary across age, ability, and region.
- **Responsible Decision-Making:** Students balance comfort, cost, and effectiveness in their redesign choices — learning to design with responsibility and purpose.
- **Collaboration & Communication:** Working in teams, they share ideas, test materials, and respectfully challenge each other's thinking to improve their design.

Engage

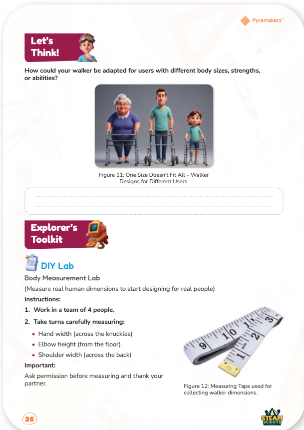
Teacher's Role:

- Facilitate empathy-based thinking by guiding students to imagine real user needs.
- Encourage inclusive design thinking using open-ended questioning.
- Offer real-world examples and challenge assumptions about "one-size-fits-all" designs.
- Support students in testing and refining their prototypes based on user diversity.
- Use visual aids, persona cards, and brief physical simulations (e.g., restricted movement) to deepen understanding.

Expected from Students:

- Identify at least one user challenge based on physical differences (size, strength, ability).
- Propose a design solution to adapt the walker for that user.
- Reflect on why the change matters and how it improves accessibility.
- Use sketches, models, or verbal explanations to share their adaptations.
- Collaborate respectfully and consider peer feedback to improve inclusiveness.

Parts of the book included.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Let's Think!

How could your walker be adapted for users with different body sizes, strengths, or abilities?

Figure 11: One Size Doesn't Fit All – Walker Designs for Different Users.

Explorer's Toolkit

DIY Lab

Body Measurement Lab

(Measure real human dimensions to start designing for real people)

Instructions:

1. Work in a team of 4 people.
2. Take turns carefully measuring:
   - Hand width (across the knuckles)
   - Elbow height (from the floor)
   - Shoulder width (across the back)

Important:

Ask permission before measuring and thank your partner.

Figure 12: Measuring Tape used for collecting walker dimensions.

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STEAM SCOUTS

Let's Think!

Objective:

Students will analyze the needs of diverse users to propose design adaptations for a walker, demonstrating their understanding of human-centered design principles. By synthesizing knowledge from previous lessons on forces and stability, they will be able to articulate how their design choices address the unique challenges of users with different body sizes, strengths, or abilities.

Possible answers:

- “We added adjustable handles so a child and an adult can both use it.”
- “We used lightweight materials so it’s easier for older people to push.”
- “We designed a wider grip so people with arthritis can hold it comfortably.”
- “We made foldable parts for storage and easier transport.”
- “We added a small seat in case someone gets tired walking.”

Possible Strategies for implementation:

1. **"See-Think-Wonder":**

Begin by presenting the image and using a "See-Think-Wonder" routine.

- **See:** Ask students what they see in the image (e.g., three people of different sizes using walkers).
- **Think:** Have them think about why the walkers might be designed differently or what problems the people might face.
- **Wonder:** Prompt them to ask questions about the design, materials, or features of the walkers. This activates their critical thinking before they even begin to answer the main question.

2. **Small Group Brainstorm:**

Have students work in small groups to brainstorm ideas for adapting the walker. This collaborative approach encourages a wider range of creative solutions. Each group can focus on one specific user from the image and discuss how the walker could be changed for them.

3. **Connecting to Previous Lessons:**

Explicitly connect this "Think" question back to earlier activities. Ask questions like: "Remember when we measured force in the 'Builder' activity? How might the force needed to push the walker change for a person with less strength?" or "How does the idea of stability from our 'DIY Lab' apply here?" This helps students see the coherence of the entire unit.

4. **Visual and Tangible Ideas:**

Encourage students to sketch their design adaptations directly on a printout of the image or on a whiteboard. They can label their new features and explain their function. This makes their abstract ideas concrete and easier for others to understand.

Explorer's Toolkit

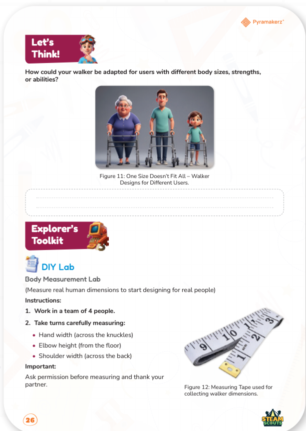
Teacher's Role:

- **Lab Supervisor:** Guides students through two distinct hands-on activities, ensuring safety and clarity in instructions.
- **Concept Clarifier:** Helps students connect their physical experiences (like pushing on a partner's hands) to the scientific principles of forces and design.
- **Facilitator:** Moves between groups, asking probing questions to stimulate critical thinking about both forces and the importance of body measurements.
- **Discussion Leader:** Leads a post-activity discussion to help students synthesize their observations and articulate their learning.

Expected From Students:

- **Active Participation:** Students must actively engage in both the partner pushing activity and the body measurement lab.
- **Careful Observation:** They are expected to pay close attention to physical sensations and accurately record data.
- **Collaboration:** Students will work respectfully with their partners and group members, taking turns and communicating effectively.
- **Critical Thinking:** They are expected to think about the "why" behind the activities and be ready to discuss their findings.

Parts of the book included.



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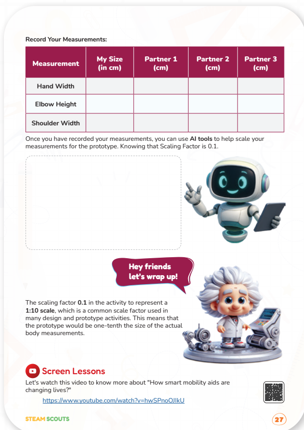

Record Your Measurements:

Measurement	My Size (in cm)	Partner 1 (cm)	Partner 2 (cm)	Partner 3 (cm)
Hand Width				
Elbow Height				
Shoulder Width				

Once you have recorded your measurements, you can use **AI tools** to help scale your measurements for the prototype. Knowing that Scaling Factor is 0.1.

Hey friends let's wrap up!

The scaling factor **0.1** in the activity to represent a **1:10 scale**, which is a common scale factor used in many design and prototype activities. This means that the prototype would be one-tenth the size of the actual body measurements.

Screen Lessons

Let's watch this video to know more about "How smart mobility aids are changing lives?"

https://www.youtube.com/watch?v=hwSPnoOJIkU

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DIY Lab

Objectives:

- Students will measure their own body dimensions (hand width, elbow height, shoulder width) and use AI tools to scale the measurements for a prototype.
- Students will understand how scaling factors can be used in design and modeling.

Materials Needed:

- Worksheets: Copies of the "DIY Lab" (partner push) and the "Body Measurement Lab" sheets for each student.
- Measurement Tools: Measuring tapes.
- Other: Pencils or pens.

Possible answers:

Sample Measurements Table

Measurement	My Size (in cm)	Partner 1 (cm)	Partner 2 (cm)	Partner 3 (cm)
Hand Width	8.5	7.8	9.2	8.0
Elbow Height	93	88	96	90
Shoulder Width	38	36	42	39

What these answers show:

- Hand Width ranges from 7.8 to 9.2 cm – small but important differences for grip and button sizes.
- Elbow Height varies due to height differences — this affects handle placement on a walker.
- Shoulder Width changes how wide a product (like a walker or strap) needs to be to fit comfortably.

Activity Instructions:

1. **Form Teams:** Divide the class into teams of four students each.
2. **Introduce the Purpose:** Explain that they will be acting as designers gathering data. Emphasize that in engineering, it's crucial to design for the human body, not just for an average size.
3. **Demonstrate Measuring:** Using a volunteer, clearly demonstrate how to take each measurement:
   - Hand width: Across the knuckles.
   - Elbow height: From the floor to the tip of the elbow.
   - Shoulder width: Across the back from shoulder to shoulder.
4. **Team Task:** Instruct the teams to carefully take turns measuring each other. Remind them to ask for permission and to thank their partners.
5. **Record Data:** Students should record their own measurements and their three partners' measurements in the provided table on their worksheet.

6. **Debrief:** After the data is collected, lead a class discussion. Ask questions like, "Why is it important to measure more than one person?" and "How could this data help you design a better walker?" This connects the raw data to the unit's overarching design challenge.

7. Once you have recorded your measurements, use AI tools to scale them for the prototype.

Note: The Scaling Factor for this activity is 0.1 (e.g., if the real measurement is 20 cm, the scaled prototype measurement will be 2 cm).

Possible Strategies for implementation:

- **"Think-Pair-Share" with the Data:** After students have collected their data, have them "Think" individually about a few questions. For example: "What was the biggest difference between your measurements and your partners'?" or "Why might a designer care about elbow height?" Then, have them "Pair" with a partner to discuss their observations before "Sharing" their findings with the larger class. This helps them move from just collecting data to analyzing its significance.
- **Color-Coded Measuring Tapes:** Use different-colored measuring tapes for each team. This makes it easy for the teacher to quickly identify which teams are working and to spot any teams that might be struggling, allowing for targeted intervention.

Explain

Teacher's Role:

- **Introduces the topics:** The teacher starts the lesson by sharing the videos and article links with the class.
- **Answers questions:** After students have watched and read, the teacher answers their questions and makes sure everyone understands the main ideas.
- **Connects the dots:** The teacher helps students see how the video about physics and the article about design all relate to the main project of designing a Smart Walker.

Expected From Students:

- **Watch and Read:** Students need to pay attention to the videos and the article.
- **Think about it:** They should be thinking about the information and how it connects to the project.
- **Join the talk:** Students should be ready to talk about what they learned and ask questions if they are confused.

parts of book included.


Screen Lessons

Let's watch this video to know more about "How smart mobility aids are changing lives?"

https://www.youtube.com/watch?v=hwSPnoOJlkU

STEAM SCOUTS

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Screen Lessons

Objective:

Students will explore how smart mobility aids, such as smart walkers and wheelchairs, are improving the lives of individuals with mobility challenges. By watching the video, students will gain an understanding of how technology is enhancing accessibility and independence for people with disabilities. This activity will help students connect technology solutions to real-world problems and consider user-centered design in their own projects.

Activity Link: https://www.youtube.com/watch?v=hwSPnoOJlkU

1. **Pre-Watching Strategies**
   - **Introduce the Topic:**
     - "What challenges might older adults face in maintaining their independence and mobility?" (Elicit responses about balance, falls, navigation, communication).
     - "What technologies already exist to assist people with mobility challenges?" (Discuss canes, walkers, wheelchairs, etc.).
   - **Set a Purpose for Viewing:**
     - "As you watch, pay close attention to the features of this 'smart' cane. What problems does it solve? How does it improve on traditional canes?"
     - "Think about who might benefit from this technology and why."
2. **During-Watching Strategies**
   - **Active Viewing Prompt:** Instruct students to jot down the key features of the smart cane as they are presented in the video.
   - **Strategic Pauses:**
     - After the smartphone replacement feature: Pause. "Why is this smartphone-like functionality important for this user group?"
     - After the fall detection feature: Pause. "How does this feature contribute to safety and peace of mind?"
     - After the GPS functionality: Pause. "What are the benefits and potential concerns related to tracking a user's location?"
     - After the cost is mentioned: Pause. "Is this price point reasonable? What factors might influence the cost and accessibility of this technology?"

- **Quick Checks:** "Give a thumbs up if you understand how that feature works."

3. **Post-Watching Strategies**

- **Feature Review:**
  - o "What are the key features of the smart cane? List them."
  - o "Which feature do you think is most important, and why?"
- **Discussion Questions:**
  - o "What are the potential benefits of this technology for users and their caregivers?"
  - o "What are some potential drawbacks or limitations of this technology?"
  - o "How might this technology evolve in the future? What other features could be added?"
  - o "Are there ethical considerations related to using this technology (e.g., privacy, dependence)?"
- **Extension Activities (Choose one or more, depending on time):**
  - o Design Challenge: "Imagine you are designing the next generation of smart mobility aids. What new features would you include, and why?"
  - o Research: "Investigate other smart mobility technologies (e.g., exoskeletons, smart walkers). How do they compare to the smart cane?"
  - o Debate: "Should technologies like this be subsidized to make them more accessible to older adults?"

Dig Deeper

Objective:

Students will learn about the importance of anthropometrics and ergonomics in design, particularly how body measurements influence the creation of products that are comfortable and functional for users. By reading the article, students will deepen their understanding of how designers use human body data to create more user-friendly products and consider these factors in their own design processes.

Article Link: https://www.technologystudent.com/designpro/ergo1.htm

Possible Strategies for implementation:

1. **Pre-Reading (Discussion)**

Strategy 1: Purposeful Prediction

Name: "Guess Before You Read"

How to Implement:

- Show the article title: "Ergonomics – Anthropometrics" and a picture of ergonomic design (e.g., chairs, keyboards).
- Ask students:
  - o "What do you think 'anthropometrics' means?"
  - o "Why might body measurements matter when making products?"

- Students can guess by:
  - Speaking in a group.
  - Sketching an idea.
  - Writing 1 quick sentence.

Teacher Language (integrating UDL + SEL): "It's okay if you're not sure! Good scientists and designers make predictions before they know all the facts."

2. During Reading

Strategy 2: Active Reading Choices

Name: "Mark, Draw, or Summarize"

How to Implement: Offer students 3 ways to engage while reading:

- Mark It: Underline or highlight any mention of body parts, sizes, or products.
- Draw It: Sketch one tool or product mentioned that uses body measurements.
- Summarize It: Write a short sentence after each paragraph answering: "Why does size matter here?"

Teacher Prompt: "You're the reader — you pick the way you stay active. Choose marking, drawing, or summarizing!"

UDL Built-In: Multiple modes of engagement (visual, kinesthetic, verbal).

SEL Built-In: Builds autonomy and responsibility:
"The way you think matters. Trust yourself to pick the method that helps you learn."

3. Post-Reading

Strategy 3: Collaborative Understanding

Name: "One Big Idea"

How to Implement: After reading, students:

- Pair up.
- Each share one big idea they learned.
- Create a Big Idea Board together:
  - Draw it.
  - Write a slogan.
  - Act it out as a mini skit.
- Options students can choose:
  - Write 1–2 sentences ("Body measurements matter because...").
  - Draw a product that was designed using anthropometrics.
  - Create a short role-play showing a badly designed vs. well-designed object.

Teacher Language: "You've gathered amazing clues. Now let's find your one big idea.

Remember, there's no one way to show smart thinking — you get to pick how you express it!"

Elaborate

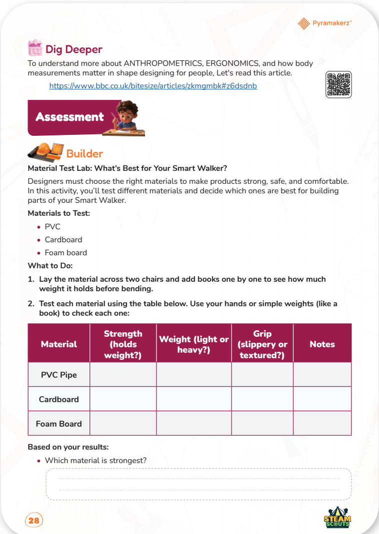
Teacher's Role:

- The teacher introduces each activity and helps students move smoothly from one task to the next.
- The teacher explains the science ideas like forces and energy, making sure students understand the videos and worksheets.
- The teacher keeps an eye on the class to make sure everyone is safe, working well together, and staying on track.
- The teacher checks student work on the "Apprentice" test and design projects to see what they have learned.

Expected from Students:

- **Students as Participants:** Students need to join in every activity, from hands-on labs to group talks and tests.
- **Students as Thinkers:** They must use the facts they learn to solve problems and think of new ideas for their Smart Walker design.
- **Students as Team Players:** They should work well with partners and in groups to complete labs and share their thoughts.
- **Students as Learners:** Students are expected to use what they learn in each part of the lesson to complete the final design challenge.

Parts of the book included.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Dig Deeper

To understand more about ANTHROPOMETRICS, ERGONOMICS, and how body measurements matter in shape designing for people, Let's read this article.

https://www.bbc.co.uk/bitesize/articles/zkmgmbk#z6dsdnb

Assessment

Builder

Material Test Lab: What's Best for Your Smart Walker?

Designers must choose the right materials to make products strong, safe, and comfortable. In this activity, you'll test different materials and decide which ones are best for building parts of your Smart Walker.

Materials to Test:

- PVC
- Cardboard
- Foam board

What to Do:

1. Lay the material across two chairs and add books one by one to see how much weight it holds before bending.
2. Test each material using the table below. Use your hands or simple weights (like a book) to check each one:

Material	Strength (holds weight?)	Weight (light or heavy?)	Grip (slippery or textured?)	Notes
PVC Pipe				
Cardboard				
Foam Board				

Based on your results:

- Which material is strongest?

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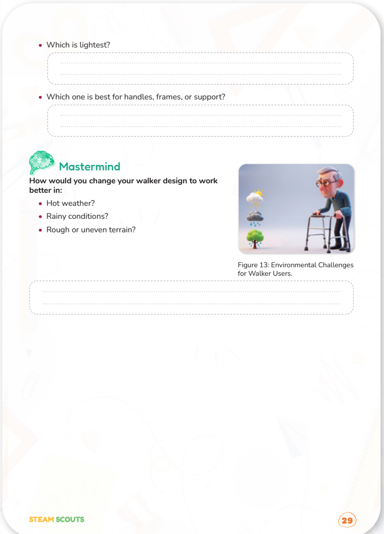
- Which is lightest?
- Which one is best for handles, frames, or support?

Mastermind

How would you change your walker design to work better in:

- Hot weather?
- Rainy conditions?
- Rough or uneven terrain?

Figure 13: Environmental Challenges for Walker Users.

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Builder

Objectives:

Students will be able to analyze and select materials for their Smart Walker based on properties like strength, weight, and grip, and propose design solutions for users with different needs and in different environments.

Needed Materials:

- Materials to Test: Cut samples of PVC pipe, cardboard, and foam board.
- Weights: A set of books or other simple, uniform weights.
- Support: Two chairs or blocks to act as a span for the materials.

Activity Instructions:

1. **Introduce the Challenge:** Explain that designers must choose the right materials to build products that are strong, safe, and comfortable. Tell students they will be testing three different materials to see which ones would be best for a Smart Walker.
2. **Part 1: Strength Test:**
    - Demonstrate how to lay one of the materials (e.g., cardboard) across two chairs or blocks.
    - Instruct students to carefully add one book at a time to the middle of the material until it bends significantly.
    - Have students record a qualitative observation of the material's strength (e.g., "Very Strong," "Weak," "Broke after 3 books") in the "Strength" column of the table.
3. **Part 2: Weight and Grip Test:**
    - Instruct students to use their hands to feel each material. They should observe and record which material feels lightest or heaviest.
    - They should also feel the texture of each material and record whether the grip is slippery or textured.
4. **Record Findings:** Remind students to fill out the "Notes" column with any other observations they have.
5. **Analyze and Conclude:** Once all tests are complete, have students individually answer the final questions on the worksheet based on their results. This will help them synthesize their findings and apply them to the Smart Walker design.

Mastermind

Objective:

Students will analyze how environmental factors such as weather and terrain affect the functionality of a walker. They will propose design modifications to improve the walker's performance in different conditions. This activity helps students engage in user-centered design, considering how external factors like weather impact mobility aids.

Activity Instructions:

In-Class Instructions:

- **Introduction to the Activity:**
  - o Introduce the activity by explaining that students will explore how weather and terrain affect the design of a walker.
  - o Present the image of a person using a walker in different weather conditions (hot weather, rainy conditions, rough terrain). Display it on the screen or distribute printed copies to the students.
  - o Explain to students that they will be considering changes to the walker design to make it better suited for these conditions.
- **Class Discussion:**
  - o Time: 15-20 minutes
  - o Lead a brief discussion about the three weather conditions: hot weather, rainy conditions, and rough terrain.
  - o Ask students to think about how each factor might affect a walker's functionality (e.g., difficulty walking on rough terrain, the walker getting wet in the rain).
  - o This is a chance to guide the students in thinking critically about design problems and potential solutions.

At-Home Instructions:

- **Design Modifications Task:**
  - o Ask students to individually think about and write down or draw the design modifications they would make to the walker for each of the three conditions (hot weather, rainy conditions, rough terrain).
  - o Encourage them to be creative and think about features that would address these environmental factors (e.g., materials for heat resistance, waterproof parts, larger wheels for uneven surfaces).
- **Optional Extension:**
  - o Students can also sketch their ideas if possible, helping them visualize their design changes.

Wrap-up and Reflection (In-Class):

- **Sharing and Reflection:**
  - o Time: 15-20 minutes (following day)
  - o After the homework is completed, have students share their ideas with the class in small groups or in a whole-class discussion.
  - o Ask students to explain why they made certain design choices based on the weather or terrain conditions.
  - o Engage the class in a discussion about how these ideas could be used in the real world and the importance of designing products that work for different environments.

Evaluate

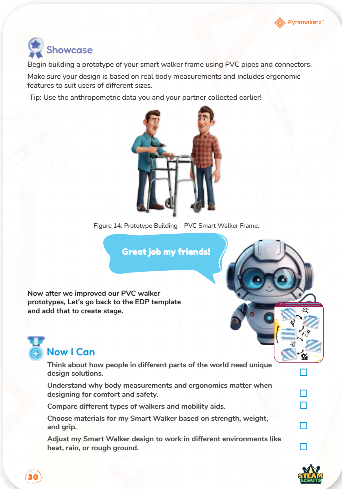
Teacher's Role:

- Guide students as they build, keeping safety in mind.
- Ask open-ended questions that help them think deeper:
- "Why did you choose that handle height?"
- "Does this width fit all the users you measured?"
- Offer feedback and encouragement as they refine their build.
- Celebrate every solution, not just the most polished ones.

Expected From Students:

- Work as a team to build a walker frame with PVC pipes and joints.
- Base the design on real user measurements collected earlier.
- Ensure the frame includes an ergonomic feature (e.g., correct height, comfort grip position, adjustable width).
- Present their prototype and explain:
    1. Who they designed for
    2. What makes it ergonomic
    3. One improvement they'd like to make

Parts of book included.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Showcase

Begin building a prototype of your smart walker frame using PVC pipes and connectors.

Make sure your design is based on real body measurements and includes ergonomic features to suit users of different sizes.

Tip: Use the anthropometric data you and your partner collected earlier!

Figure 14: Prototype Building – PVC Smart Walker Frame.

Great job my friends!

Now after we improved our PVC walker prototypes, Let's go back to the EDP template and add that to create stage.

Now I Can

- Think about how people in different parts of the world need unique design solutions. ☐
- Understand why body measurements and ergonomics matter when designing for comfort and safety. ☐
- Compare different types of walkers and mobility aids. ☐
- Choose materials for my Smart Walker based on strength, weight, and grip. ☐
- Adjust my Smart Walker design to work in different environments like heat, rain, or rough ground. ☐

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Showcase

Objective:

Students will build a prototype of a smart walker frame using PVC pipes and connectors, applying real body measurements and ergonomic design principles to create a functional and comfortable mobility aid. By using anthropometric data, students will ensure their design accommodates users of different sizes, reinforcing the importance of user-centered design in engineering projects.

Materials needed:

- PVC pipes (various lengths)
- Elbow and T-joints
- Tape measure or ruler
- Student-collected anthropometric data
- Sketch paper and pencils (optional for planning)
- Optional: rubber caps or foam for handle placeholder

Resources link: https://docs.google.com/document/d/1Nlehpz4EOkbZlVrAS6urj0gfajqHCfHA/edit?usp=drive_link&ouid=110807401694106266915&rtpof=true&sd=true

https://docs.google.com/document/d/1qPjYd8ev_SgX9GzZYzPw3mMUOT1oC2LU/edit?usp=sharing&ouid=103342660162119036856&rtpof=true&sd=true

Step by step instructions:

1. **First: Before You Start: Planning**
   - **Review Your User Measurements:**
     - Look at the hand height, shoulder width, and elbow height of your target users (from previous activity).
     - Decide who your walker is for: Tall person? Short person? Can it fit both?
   - **Sketch Your Frame Design (Optional but Recommended):**
     - Draw what your walker might look like.
     - Plan how high the handles should be and how wide the frame will be.
2. **Second: Build Instructions**

Step 1: Build the Base (Bottom Frame)

- Take two longer PVC pipes to form the sides of the base (the bottom where the walker touches the floor).
- Connect them with two shorter pipes using T-joints to form a rectangle.
- Make sure the frame is flat and stable. This will support the vertical parts.

Tip: The walker base should be slightly wider than the average shoulder width of your user.

Step 2: Add the Upright Posts (Vertical Supports)

- Insert vertical PVC pipes into the upward-facing part of each T-joint.
- These pipes will form the legs of the walker — connecting the base to the handles.
- Use your ruler to measure the height from the ground to your user's elbow height (while sitting or standing). That's the target height for the handles.

Tip: Use two different heights if you're designing for users with different needs (e.g., adjustable concept).

Step 3: Add the Top Bar (Handlebar Area)

- Use PVC elbow joints to connect a horizontal pipe across the two vertical legs.
- This becomes the handlebar.
- You can add T-joints or extra horizontal bars if you want two handles instead of one.

Optional: Wrap the handlebar with foam or cloth to simulate a soft grip.

Step 4: Check for Ergonomics and Comfort

- Ask a classmate to act as the "user."
- Have them hold the walker like they would if they were walking with it.
- Check:
  - Are the handles at a comfortable height?
  - Is the width enough for their body?
  - Are their arms relaxed and slightly bent at the elbow?

Possible Strategies for implementation:

Strategy 1: Design-by-Data

Implementation:

Before building, ask students to:

- Look at their team's anthropometric measurements.
- Choose at least two users to design for (e.g., one tall, one short).
- Sketch or plan handle height and frame width accordingly.

Teacher Prompt: "Your walker isn't for just one person — it should work for different people. How can your frame adjust or adapt to their sizes?"

Strategy 2: Prototype & Iterate

Implementation:

- Teams assemble their PVC walker frame based on their plan.
- Test their frame with different classmates as mock users.
- Reflect and make adjustments (height, spacing, handle width, etc.).

Teacher Prompt: "Let's treat this as version 1. What works? What doesn't? What would you do differently next round?"

Strategy 3: Design Justification

Implementation:

- Once frames are built, each team shares:
  - What user measurements they based their design on
  - What design feature they added to increase comfort or safety
  - One thing they would still improve

Teacher Prompt: "Tell us your team's thinking. Every design has a story — what was yours?"

Reflection Questions:

- Did you design for more than one type of user?
- Did you test with real measurements?
- Did you solve a real-world problem?
- "If someone your grandparents' age tried your walker, how would they feel using it? What's one thing you'd change to make it even better for them?"
  Let students journal, discuss in groups, or draw changes they'd make next.

Rubric:

Criteria	4-Excellent	3-Good	2-Developing	1-Needs Support
Ergonomic Fit	Design fits multiple users using real measurements with strong justification	Design fits one user well and considers comfort	Design fit is somewhat accurate, minimal adjustment	Design doesn't reflect user measurements
Use of Materials	PVC pieces used creatively and effectively for safe, balanced design	Most PVC joints used appropriately, structure is stable	Some difficulty in assembly; design is wobbly or uneven	Structure is unstable or not completed
Team Collaboration	Team worked with strong communication, sharing roles equally	Team mostly worked well, shared tasks	Uneven roles or some disagreement	Team had trouble cooperating or completing together
Presentation & Explanation	Clearly explained design purpose, users, and possible improvements	Explained most parts clearly with minor gaps	Struggled to explain design choices	Gave little or unclear explanation
Creativity & Problem-Solving	Design includes innovative thinking or extra features	Design is thoughtful and shows effort	Basic solution with limited adjustments	Shows little creativity or design thinking

Don't forget to go back to the EDP page and add these to create part.

Create

- We built our Smart Walker frame prototype using PVC pipes and joints based on real anthropometric data that we measured.
- Our walker includes ergonomic features to fit users of varying sizes.

Now I Can

Objective:

By the end of this self-reflection activity, students will be able to evaluate their learning about inclusive design and ergonomics by reflecting on how their Smart Walker design choices support safety, comfort, and usability in different environments and for different users.

Activity instructions:

- Ask students to read each "Now I Can" statement silently.
- Give them quiet time to think about evidence from their design (sketches, models, or discussions).
- Encourage them to connect each statement to something they actually did during the lesson.
- Ask students to point to or circle parts of their design (digital model or drawing) that match each statement.
- This supports students who struggle with written reflection.

Relation to Standards

1. Next Generation Science Standards (NGSS)

Relevant Performance Expectations:

- MS-PS2-1: Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects.
- MS-ETS1-1: Define a design problem that reflects a need or want and includes criteria for success and constraints.
- MS-ETS1-2: Evaluate competing design solutions using a systematic process.

How the Lesson Aligns:

- Students explore and apply Newton's Third Law through hands-on experiments and force analysis.
- They engage in an engineering design challenge by creating a smart walker prototype based on real data.
- They test and revise their prototype based on ergonomics and user feedback, fulfilling design process requirements.

2. International Baccalaureate (IB) – MYP Design & Sciences

MYP Sciences Criteria:

- Criterion A: Knowing and understanding
- Criterion B: Inquiring and designing
- Criterion C: Processing and evaluating
- Criterion D: Reflecting on the impacts of science

MYP Design Criteria:

- Inquiring and analyzing real human needs (anthropometrics, inclusive design)
- Developing ideas through sketching and planning
- Creating the solution using tools and materials (PVC walker prototype)
- Evaluating based on performance and user comfort

How the Lesson Aligns:

- Students investigate a real-world problem (mobility challenges for the elderly).
- They go through all stages of the design cycle, from identifying user needs to creating and improving a physical product.

3. UNESCO Education for Sustainable Development Goals (SDGs)

- Goal 3: Good Health and Well-being
- Goal 4: Quality Education
- Goal 9: Industry, Innovation, and Infrastructure
- Goal 10: Reduced Inequalities

How the Lesson Aligns:

- The project focuses on designing inclusive, health-supportive technologies (walkers for people with mobility issues).
- It encourages innovation, empathy, and design thinking — all key for solving real societal issues.
- Students are taught to consider diversity in body types and physical needs, promoting inclusion.

4. ISTE Standards for Students (Technology and Innovation)

- Innovative Designer: Students use a design process to identify and solve problems.
- Computational Thinker: Students collect and analyze data (anthropometrics) to inform decisions.
- Empowered Learner: Students take ownership of learning through exploration and creativity.
- Creative Communicator: Students express understanding through sketches, models, and presentations.

How the Lesson Aligns:

- Students collect real data, build prototypes, and present design decisions, all while solving a practical problem.
- The process fosters collaboration, creativity, and ownership over learning.

5. CASEL – Social and Emotional Learning (SEL)

- Self-awareness: Understanding their own learning styles and design thinking abilities.
- Social awareness: Designing for others with empathy.
- Relationship skills: Collaborating, sharing feedback, and co-building.
- Responsible decision-making: Designing inclusive, safe solutions.